

CSE200: Computer Aided Engineering Drawing

Tahsin Tariq Banna

Lecturer,
Department of Computer Science and Engineering,
East West University

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Sectional Views

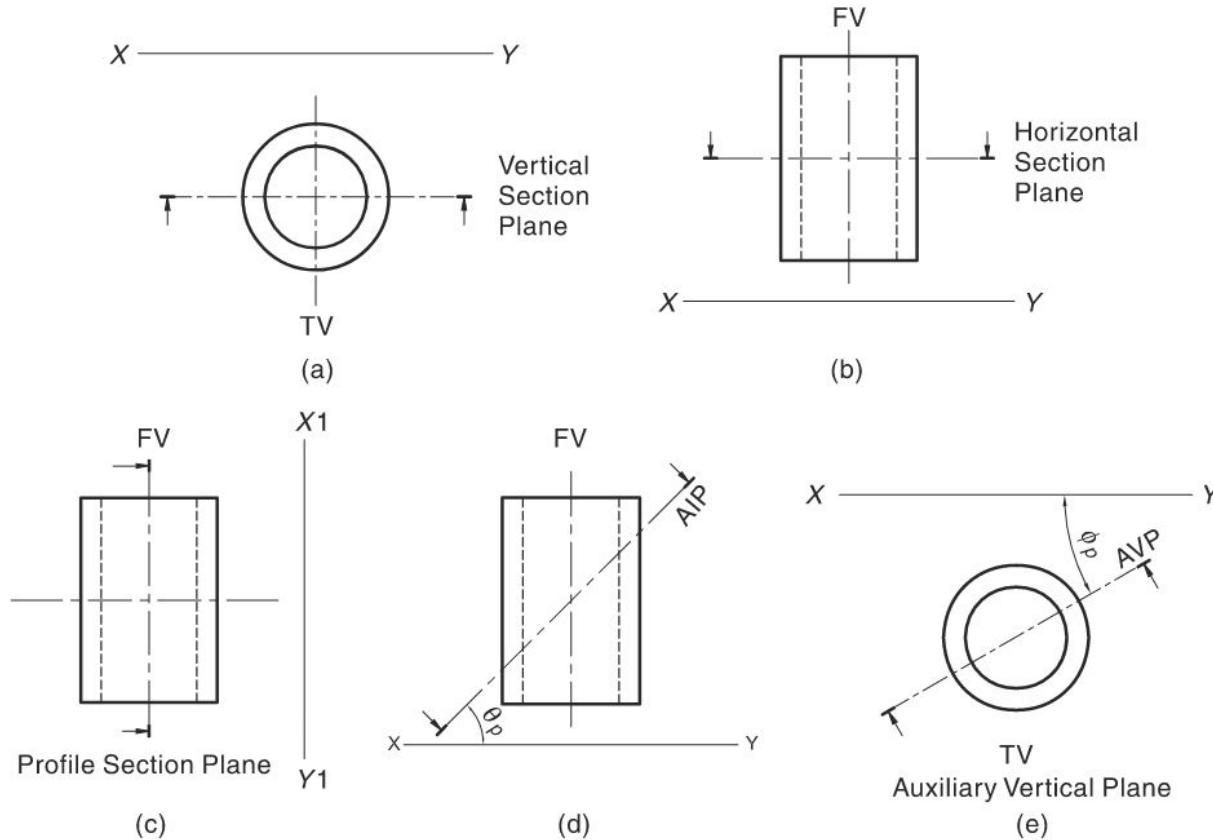
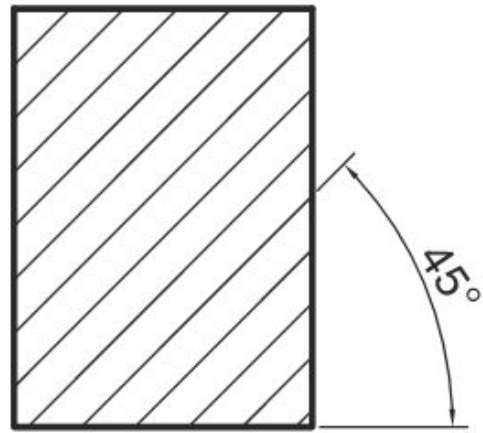
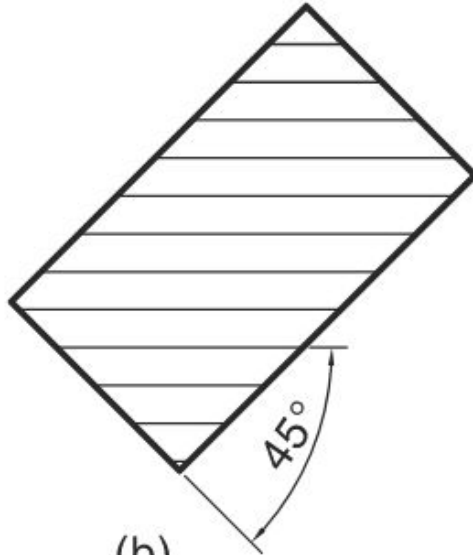


Fig. 9.25 Types of cutting planes and their representation

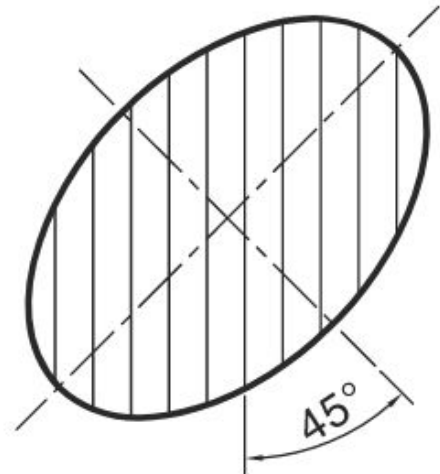
Sectional Views



(a)



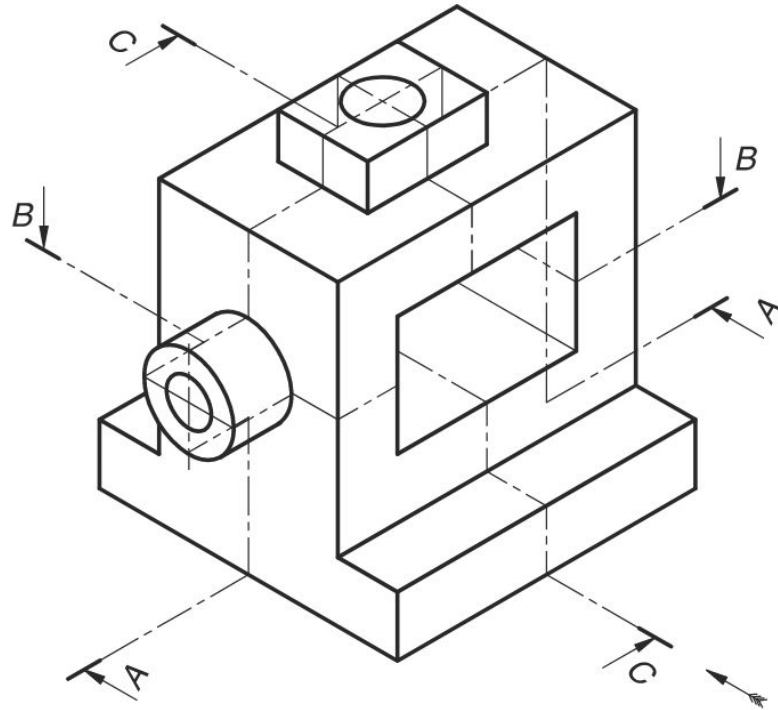
(b)



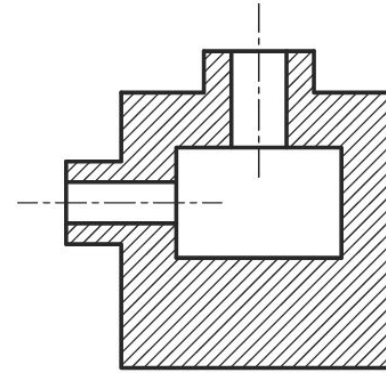
(c)

Fig. 9.26

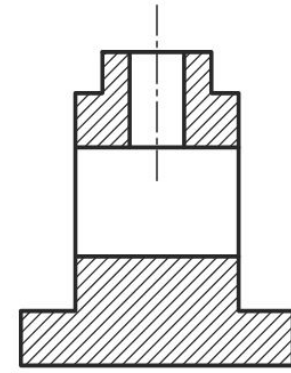
Sectional Views



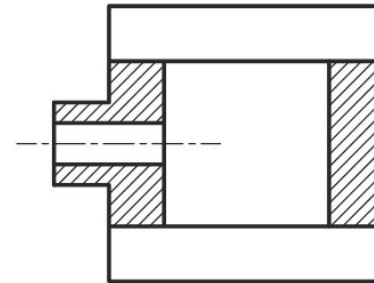
(a)



SECTIONAL FV
(Section A-A)



SECTIONAL SV
(Section C-C)



SECTIONAL TV
(Section B-B)

(b)

Fig. 9.27

Methods of Sectioning

There are different ways of sectioning the object. The object is sectioned by considering its features. The sectioning should be made in such a way that all the complicated internal features of the object will be as clear as possible. The various methods of sectioning are explained below.

- **Full Section** The sectional view obtained after removing the front-half portion of an object through its centre is known as a full section
- **Half Section** The sectional view obtained after removing the front quarter portion by means of two cutting planes at right angles to each other is known as half section
- **Offset Section** The sectional view obtained by a cutting plane in a zigzag way so as to reveal the maximum details of the object is known as an offset section

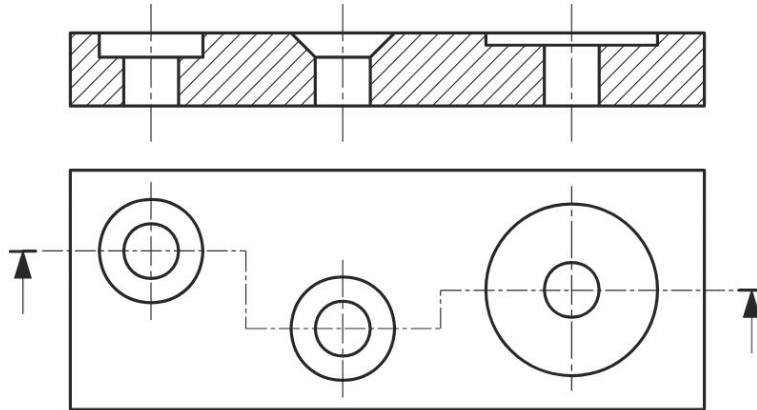


Fig. 9.29

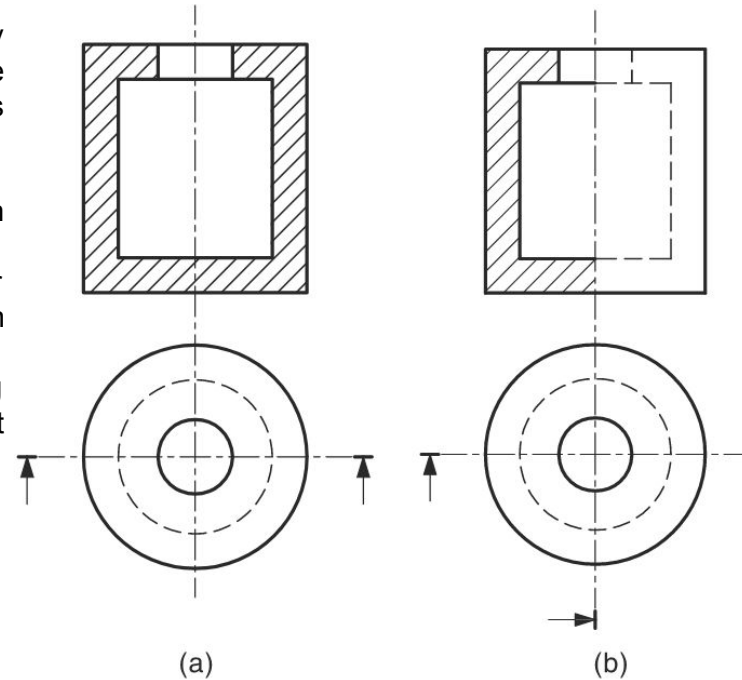


Fig. 9.28

Methods of Sectioning

- **Revolved Section** A revolved section is used to show the uniform shape of the object from end to end, Fig. 9.30.
- **Removed Section** A removed section is used to show the variable shape of the object from end to end, Fig. 9.31.
- **Aligned Section** An aligned section is used to show the shape of features that do not align with the vertical and horizontal centrelines of the object, Fig. 9.32.

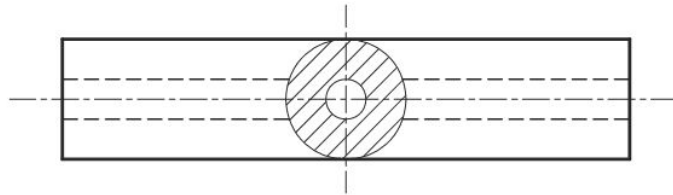


Fig. 9.30

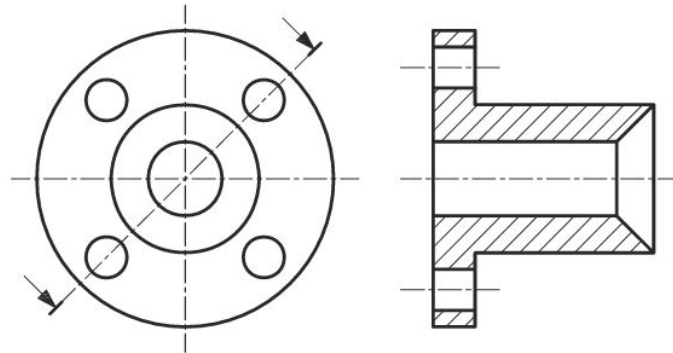


Fig. 9.32

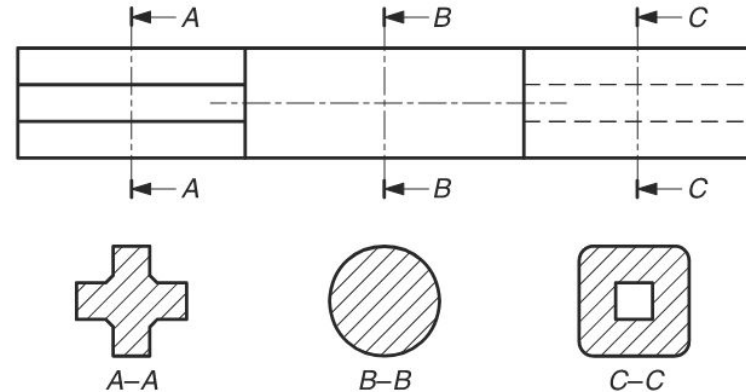


Fig. 9.31

Methods of Sectioning

- **Ribs and Other Standard Parts in Section** When a cutting plane passes longitudinally through the centre of the ribs, spokes, webs or other standard parts, they are not shown sectional because it gives a wrong impression of the thickness or of the other details, Fig. 9.33.

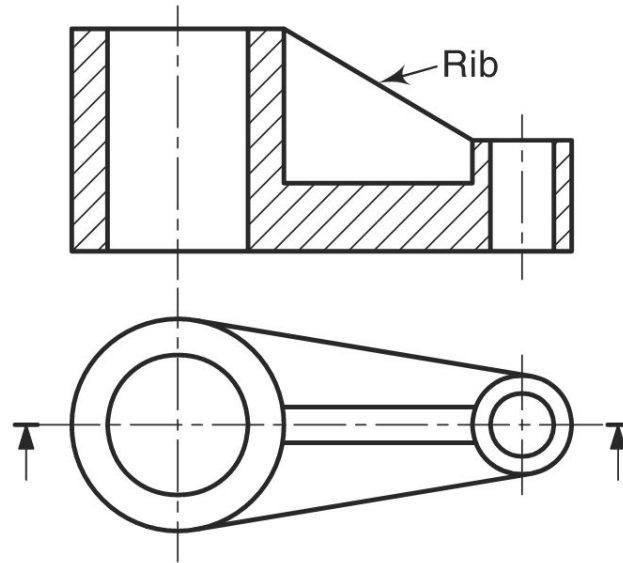
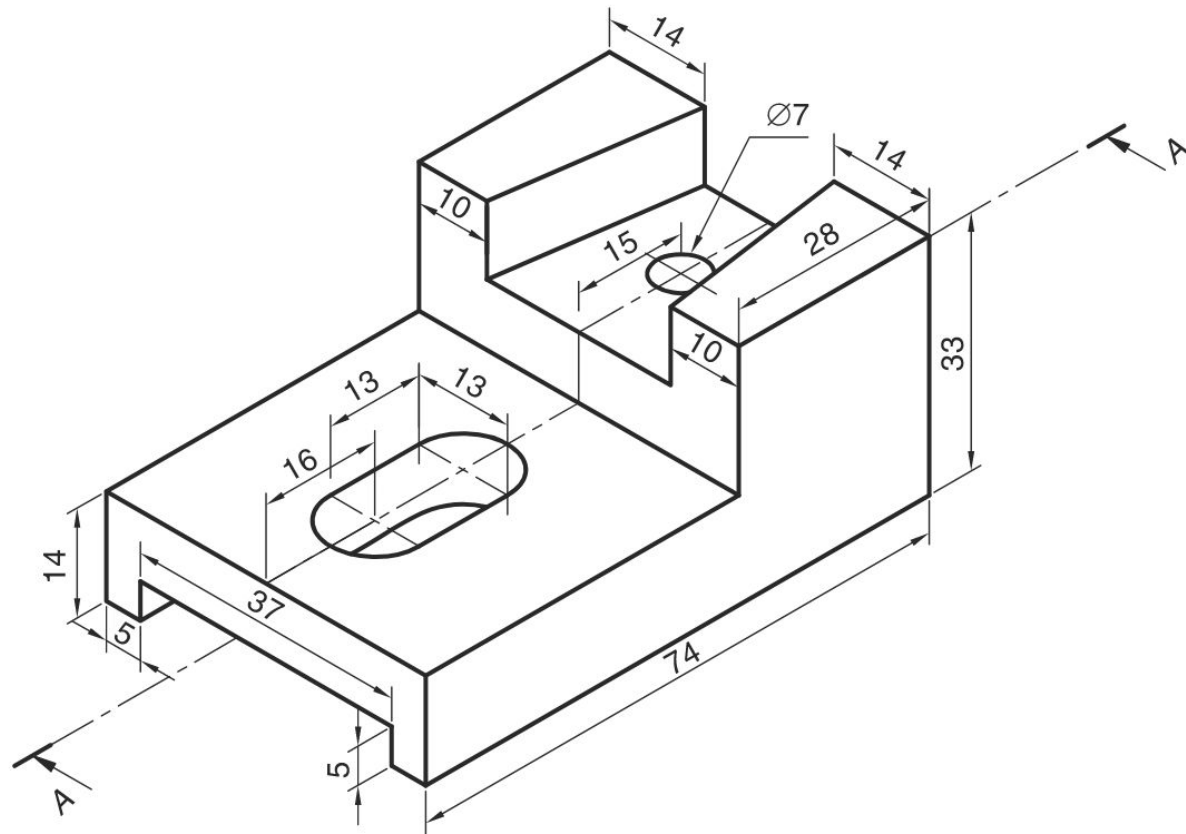
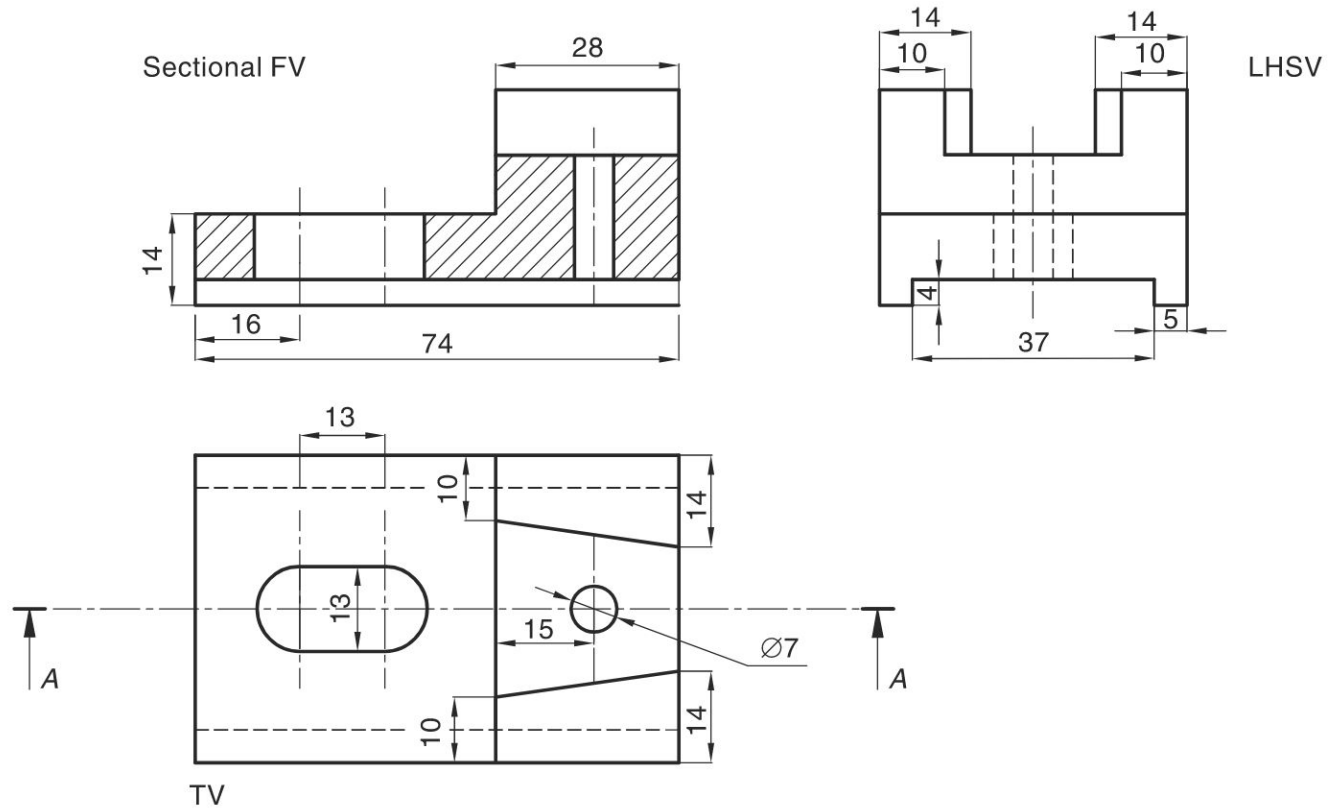


Fig. 9.33

Methods of Sectioning



Methods of Sectioning



(b)

Methods of Sectioning

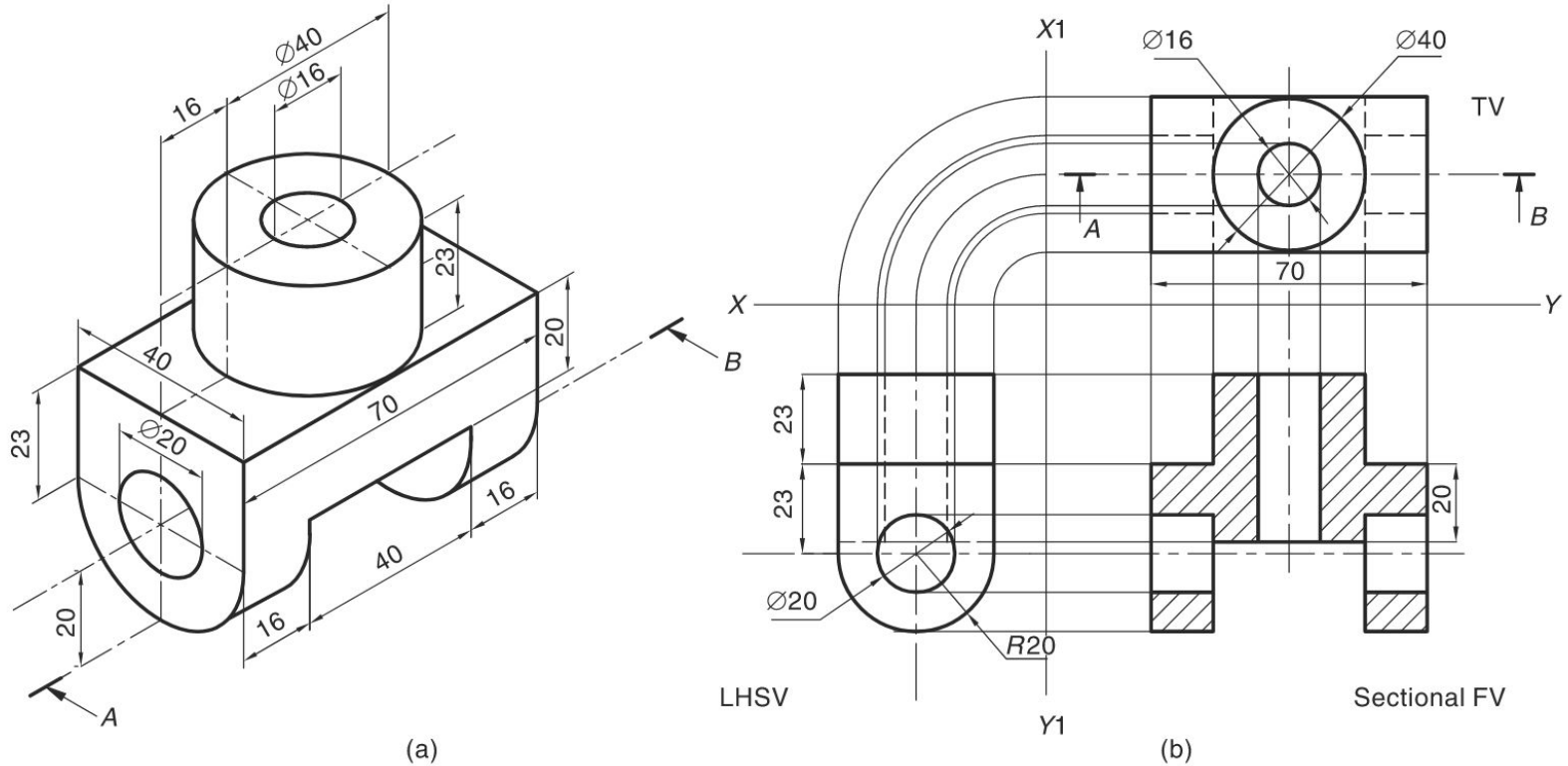
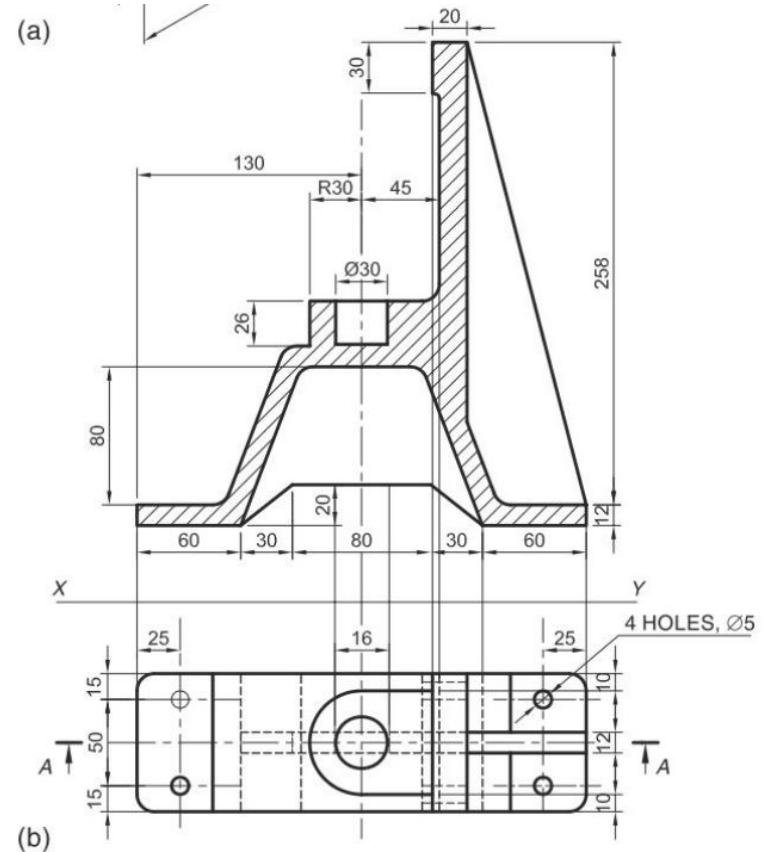
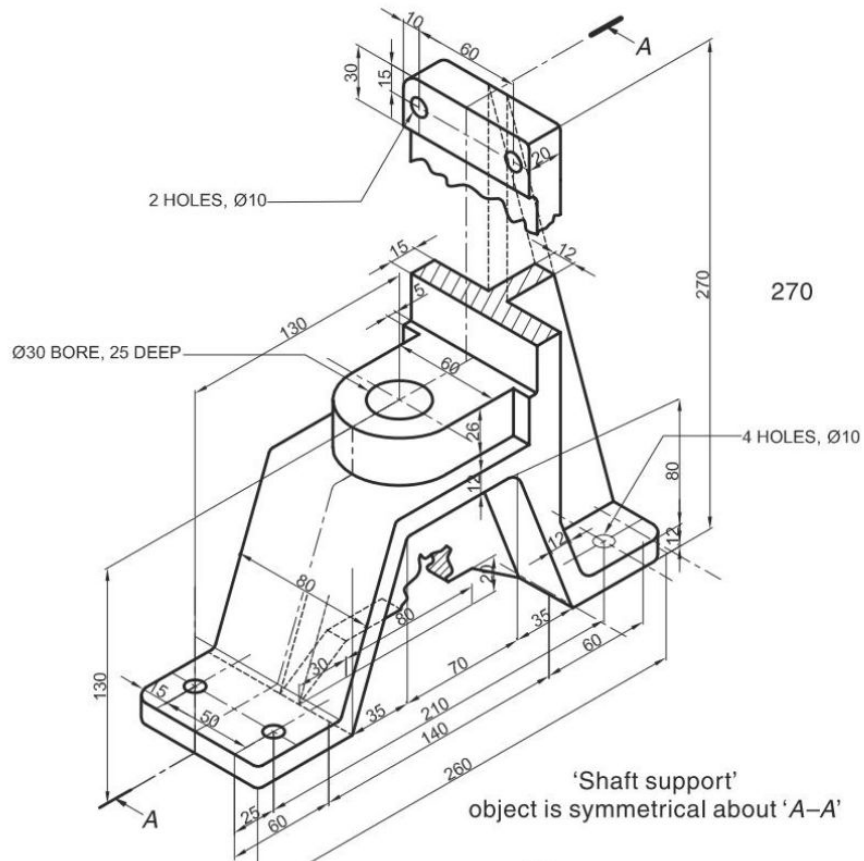
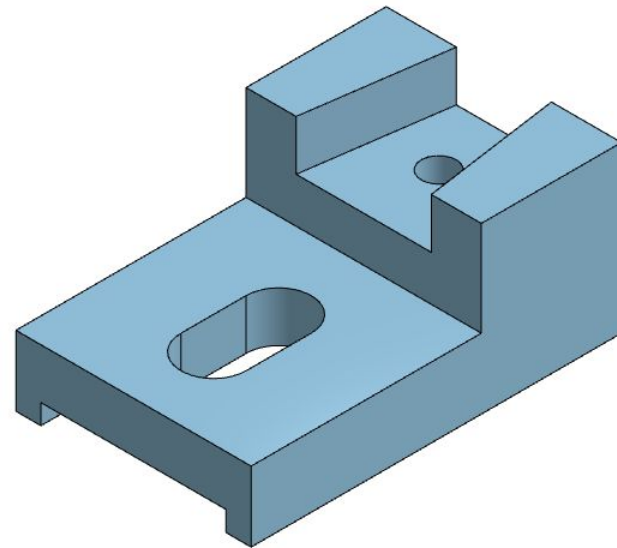
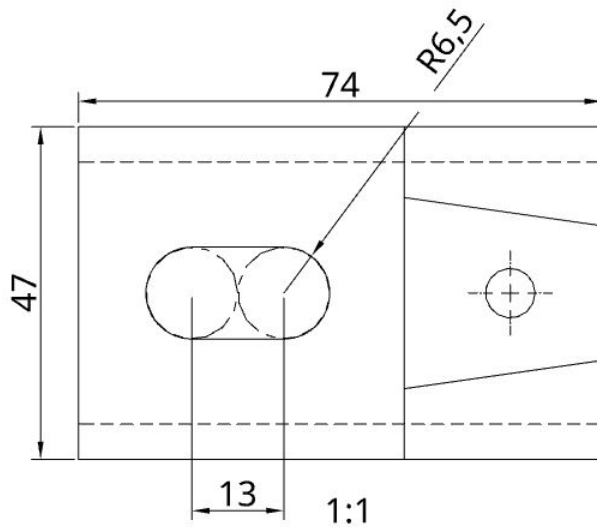
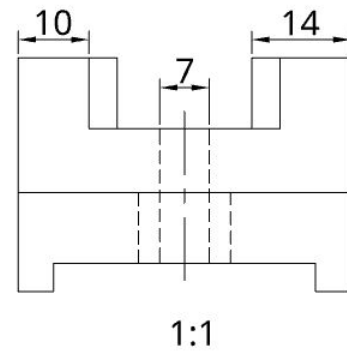
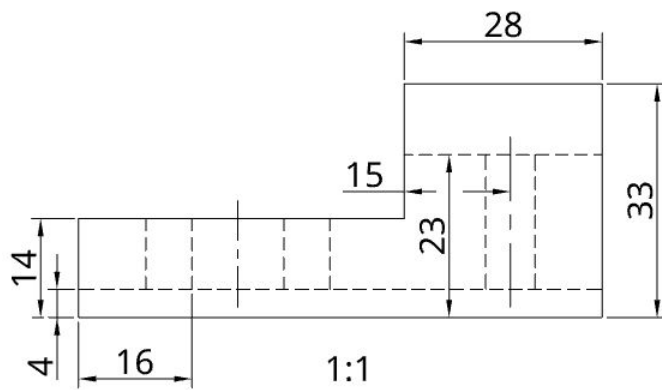


Fig. 9.35

Methods of Sectioning



Onshape



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